

Notes on the Breidbart Basis

Intercept-and-Resend Attack on BB84 and
Application to the IQT1 Quantum Token Protocol

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Contents

1	The BB84 Bases	2
2	The Breidbart Basis	2
2.1	Geometric interpretation	2
3	Intercept-and-Resend Attack in the Breidbart Basis	2
3.1	Setting up the attack	2
3.2	Computing the success probability	2
3.3	Why $\pi/8$ is optimal	3
4	QBER in the Two Scenarios	3
4.1	QKD scenario (Eve re-sends a photon)	3
4.2	IQT scenario (Eve reports outcomes directly)	4
5	Mutual Information	4
5.1	Eve's information about the correct bit	4
5.2	Alice–Bob channel	4
6	Application to the IQT1 Protocol	4
6.1	Protocol structure	4
6.2	QBER observed by the bank	5
6.3	Bank's acceptance criterion	5
6.4	Unforgeability condition and Q_{tol}	5
7	Unforgeability Bound: Equation (D1)	6
7.1	Connection to the Breidbart attack	6
7.2	Simulation confirmation	6
8	Summary	6

1 The BB84 Bases

The BB84 protocol uses two orthonormal bases of the qubit space $\mathbb{C}^2$:

$$\text{Computational basis } (Z) : \quad |0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1)$$

$$\text{Hadamard basis } (X) : \quad |+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (2)$$

The four BB84 states are $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$. On the Bloch sphere, the Z basis lies along the z -axis and the X basis lies along the x -axis; they are separated by an angle of 90° on the sphere (equivalently, 45° in the Hilbert space picture).

2 The Breidbart Basis

Definition 2.1 (Breidbart basis):

The *Breidbart basis* $\mathcal{B} = \{|\phi_0\rangle, |\phi_1\rangle\}$ is defined by

$$|\phi_0\rangle = \cos\frac{\pi}{8} |0\rangle + \sin\frac{\pi}{8} |1\rangle \quad (3)$$

$$|\phi_1\rangle = -\sin\frac{\pi}{8} |0\rangle + \cos\frac{\pi}{8} |1\rangle \quad (4)$$

It is an orthonormal basis since $\langle\phi_0|\phi_1\rangle = 0$ and $\| |\phi_0\rangle \| = \| |\phi_1\rangle \| = 1$.

2.1 Geometric interpretation

The Breidbart basis is obtained by rotating the Z basis by $\pi/8 = 22.5^\circ$ in the Hilbert space, which corresponds to a rotation of $\pi/4 = 45^\circ$ on the Bloch sphere. This places $\mathcal{B}$ *exactly halfway* between the Z and X bases, as illustrated below:

Basis	Bloch-sphere angle from Z	Hilbert-space angle from Z
Z	0°	0°
Breidbart	45°	22.5°
X	90°	45°

This midpoint position is precisely what makes $\mathcal{B}$ the optimal intercept-and-resend basis.

3 Intercept-and-Resend Attack in the Breidbart Basis

3.1 Setting up the attack

Bob (the bank, sender) prepares one of the four BB84 states and sends it as a single photon to Alice. Eve (= dishonest Alice in the IQT context) intercepts every photon, measures it in the Breidbart basis $\mathcal{B}$, and then either (i) re-sends the corresponding Breidbart state to Bob's agent, or (ii) in the IQT context, records her outcome and reports it as her token bit.

3.2 Computing the success probability

We compute $P(\text{Eve correct})$ for each BB84 state.

Z basis states.

$$|\langle \phi_0 | 0 \rangle|^2 = \cos^2 \frac{\pi}{8} = \frac{2 + \sqrt{2}}{4} = \frac{1}{2} + \frac{1}{2\sqrt{2}} \quad (5)$$

$$|\langle \phi_1 | 0 \rangle|^2 = \sin^2 \frac{\pi}{8} = \frac{2 - \sqrt{2}}{4} = \frac{1}{2} - \frac{1}{2\sqrt{2}} \quad (6)$$

So Eve's probability of identifying $|0\rangle$ correctly is $\cos^2(\pi/8) = \frac{1}{2} + \frac{1}{2\sqrt{2}}$. By symmetry, the same holds for $|1\rangle$.

X basis states.

$$|\langle + | \phi_0 \rangle|^2 = \left(\frac{\cos \frac{\pi}{8} + \sin \frac{\pi}{8}}{\sqrt{2}} \right)^2 = \frac{1}{2} (1 + 2 \cos \frac{\pi}{8} \sin \frac{\pi}{8}) = \frac{1}{2} (1 + \sin \frac{\pi}{4}) = \frac{1}{2} + \frac{1}{2\sqrt{2}} \quad (7)$$

Again, the same value. By symmetry this holds for $|-\rangle$ as well.

Result 3.1 (Breidbart success probability):

For *every* BB84 state, Eve's probability of identifying the correct bit when measuring in the Breidbart basis is

$$p_{BE} = \frac{1}{2} + \frac{1}{2\sqrt{2}} \approx 0.8536 \quad (8)$$

and the error probability is

$$e_{BE} = 1 - p_{BE} = \frac{1}{2} - \frac{1}{2\sqrt{2}} \approx 0.1464 \quad (9)$$

This is independent of which BB84 state was sent and which basis Bob used.

3.3 Why $\pi/8$ is optimal

Consider the general rotated basis $\mathcal{B}_\theta$ with angle $\theta/2$ in Hilbert space. The overall success probability (averaging over both BB84 bases with equal probability) is

$$p_{\text{right}}(\theta) = \frac{1}{2} \cos^2 \frac{\theta}{2} + \frac{1}{4} [1 + \sin \theta] \quad (10)$$

Taking the derivative and setting to zero:

$$\frac{dp_{\text{right}}}{d\theta} = \frac{1}{4} (\cos \theta - \sin \theta) = 0 \implies \theta = \frac{\pi}{4} \quad (11)$$

which corresponds to Hilbert-space angle $\pi/8$, confirming that the Breidbart basis is the unique optimal intercept-and-resend basis for BB84.

4 QBER in the Two Scenarios

The error rate Bob observes depends critically on the scenario.

4.1 QKD scenario (Eve re-sends a photon)

In standard QKD, after measuring in $\mathcal{B}$, Eve re-sends the corresponding Breidbart state $|\phi_i\rangle$ to Bob. Bob then measures in his correct BB84 basis. The probability that Alice and Bob *agree* is

$$p_{AB} = p_{BE}^2 + e_{BE}^2 = \left(\frac{1}{2} + \frac{1}{2\sqrt{2}} \right)^2 + \left(\frac{1}{2} - \frac{1}{2\sqrt{2}} \right)^2 = \frac{3}{4} \quad (12)$$

giving a **QBER** of $Q = 1 - p_{AB} = 0.25$.

4.2 IQT scenario (Eve reports outcomes directly)

In the IQT1 protocol, Eve (= dishonest Alice) does *not* re-send a photon. Instead, she records her Breidbart outcome x_k and later reports it as her token bit. Bob (the bank) checks x_k against the prepared bit t_k on the subset Δ_b . The error rate is simply

$$\text{QBER}_{\text{IQT}} = P(x_k \neq t_k) = e_{BE} = \frac{1}{2} - \frac{1}{2\sqrt{2}} \approx 0.1464 \quad (13)$$

Scenario	QBER	Value
QKD (Eve re-sends photon)	$1 - (p_{BE}^2 + e_{BE}^2)$	0.25
IQT (Eve reports directly)	e_{BE}	0.1464

5 Mutual Information

5.1 Eve's information about the correct bit

Eve's channel to the correct bit t_k is a binary symmetric channel (BSC) with crossover probability e_{BE} . Her mutual information per bit is

$$I_{BE} = 1 - h(e_{BE}) = 1 - h\left(\frac{1}{2} - \frac{1}{2\sqrt{2}}\right) \approx 0.399 \text{ bits} \quad (14)$$

where $h(q) = -q \log_2 q - (1 - q) \log_2 (1 - q)$ is the binary entropy.

5.2 Alice–Bob channel

In QKD, the channel between Alice (sender) and Bob (receiver) after the attack is a BSC with crossover $Q = 0.25$, giving

$$I_{AB} = 1 - h(0.25) \approx 0.189 \text{ bits} \quad (15)$$

The secret key rate (individual attack, Csiszár–Körner) is

$$k = \max(I_{AB} - I_{BE}, 0) = \max(0.189 - 0.399, 0) = 0 \quad (16)$$

No secret key can be extracted; the full Breidbart attack is detectable in QKD.

6 Application to the IQT1 Protocol

6.1 Protocol structure

In IQT1 (Kent et al. [1]), Bob sends N BB84 single photons to Alice. Alice measures, commits to outcomes, and later presents a token at one of two spacelike-separated points Q_0 or Q_1 . The key structure is:

- Bob's protocol algebra ensures $\tilde{d}_{b,k} = y_k$ (Alice's reported basis) always.
- Therefore $\Delta_b = \{k : y_k = u_k\}$, the positions where Alice *claimed* to use Bob's basis.
- Since u_k is uniform and y_k is independent (for the Breidbart attack), $|\Delta_0| \approx |\Delta_1| \approx N/2$.

6.2 QBER observed by the bank

At presentation point Q_b , Bob checks $x[\Delta_b]$ against $t[\Delta_b]$. The observed QBER is

$$\text{QBER}_b = e_{BE} = \frac{1}{2} - \frac{1}{2\sqrt{2}} \approx 14.64\% \quad (17)$$

for both $b = 0$ and $b = 1$, by the independence of the Breidbart outcome from the reported basis.

6.3 Bank's acceptance criterion

The bank sets a threshold γ_{err} (a *protocol parameter*, chosen before the protocol runs). It accepts the token if and only if $\text{QBER}_b \leq \gamma_{\text{err}}$.

For a perfect channel (honest Alice has $\text{QBER} = 0$), any choice of $\gamma_{\text{err}} \in (0, e_{BE})$ gives perfect separation: honest Alice is always accepted (her QBER is identically zero) and the Breidbart attack is always detected (Eve's QBER $e_{BE} \approx 14.64\%$ exceeds every such γ_{err}). In practice one sets $\gamma_{\text{err}} \approx 5\%$, well below $e_{BE} = 14.64\%$.

6.4 Unforgeability condition and Q_{tol}

For the information-theoretic analysis, we ask: for what QBER q does Eve have enough mutual information about the *reciprocal* sub-token T_{1-b} (size $n_Z \approx N/2$) to potentially forge it?

Eve's total mutual information over N bits is $(1 - h(e_{BE})) \cdot N$. This is split equally between the two sub-tokens:

$$\text{Info about } T_{1-b} = \frac{1 - h(q)}{2} \cdot N \quad \text{bits} \quad (18)$$

Eve can forge if this exceeds the number of correct bits she needs:

$$\text{Eve can forge} \iff \frac{1 - h(q)}{2} > q \quad (\text{Condition 1})$$

The *maximum tolerable QBER* is defined as the crossover point:

Result 6.1 (Maximum tolerable QBER):

$$Q_{\text{tol}} \text{ is the solution of } \frac{1}{2}(1 - h(q)) = q \implies \boxed{Q_{\text{tol}} \approx 17.05\%} \quad (19)$$

The factor $\frac{1}{2}$ comes from the equal split of Eve's N qubits between the two sub-tokens Δ_0 and Δ_1 .

Remark 6.1 (Two distinct thresholds):

γ_{err} and Q_{tol} play different roles:

- $\gamma_{\text{err}} \approx 5\%$ is **chosen** by the bank as a protocol parameter. It is the acceptance threshold in the actual check.
- $Q_{\text{tol}} \approx 17.05\%$ is **calculated** from information theory. It is the boundary of the forgeable region.

The Breidbart QBER satisfies $\gamma_{\text{err}} < e_{BE} < Q_{\text{tol}}$, so the attack fails at the γ_{err} check before the information-theoretic bound is needed.

7 Unforgeability Bound: Equation (D1)

Result 7.1 (Lemma 1 / eq. (D1), Kent et al. [1]):

The ideal quantum token schemes IQT1 and IQT2 are ε_{unf} -unforgeable with

$$\varepsilon_{\text{unf}} = \left(\frac{1}{2} + \frac{1}{2\sqrt{2}} \right)^N = p_{BE}^N \quad (\text{D1})$$

7.1 Connection to the Breidbart attack

The base of eq. (D1) is *exactly* the Breidbart per-qubit success probability p_{BE} . This is not a coincidence: the proof of Lemma 1 in Appendix D of [1] reduces the general cheating strategy to a state-discrimination task, and the optimal strategy for that task is precisely the Breidbart measurement.

The bound can be understood as: over N independent qubits, Eve's joint probability of guessing all bits correctly is at most p_{BE}^N , which decays exponentially. Selected values:

N	$\varepsilon_{\text{unf}} = p_{BE}^N$
100	$\approx 10^{-7}$
500	$\approx 10^{-34}$
1000	$\approx 10^{-69}$
2000	$\approx 10^{-138}$

7.2 Simulation confirmation

A direct simulation of the IQT1 protocol (single photon, perfect channel, $N_{\text{trials}} = 500$ per point) confirms:

- For $N \leq 200$: the simulated P_{forge} is positive and consistent with p_{BE}^N .
- For $N \geq 300$: zero forging events are observed in 500 trials, consistent with $\varepsilon_{\text{unf}} \leq 10^{-21}$.
- The simulated QBER $\hat{e}_{BE} = 0.1462 \pm 0.0117$ matches the theoretical $e_{BE} = 0.1464$ to within one standard deviation.

8 Summary

Quantity	Value	Nature
$p_{BE} = \frac{1}{2} + \frac{1}{2\sqrt{2}}$	0.8536	calculated (Born rule)
$e_{BE} = \frac{1}{2} - \frac{1}{2\sqrt{2}}$	0.1464	calculated (IQT QBER)
γ_{err}	$\sim 5\%$	chosen (protocol parameter)
Q_{tol}	17.05%	calculated (info theory)
$\varepsilon_{\text{unf}} = p_{BE}^N$	$\sim 10^{-69}$ at $N = 1000$	calculated (eq. D1)

The chain of logic for a single-photon, perfect-channel scenario:

$$\underbrace{\gamma_{\text{err}}}_{\text{chosen}} < \underbrace{e_{BE}}_{\text{QBER}} < \underbrace{Q_{\text{tol}}}_{\text{info bound}} \implies \text{Breidbart attack always detected, } \varepsilon_{\text{unf}} \rightarrow 0 \quad (20)$$

References

- [1] A. Kent, D. Lowndes, D. Pitalúa-García, J. Rarity, *Practical quantum tokens without quantum memories and experimental tests*, arXiv:2104.11717v4 (2022).